

AYUSH TIWARI

AI/ML Engineer | Python | TensorFlow | Computer Vision | Full-Stack Developer

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Summary

Final-year B.Tech (AI & ML) student with 2+ years of hands-on experience in Machine Learning, Deep Learning, Computer Vision, and Full-Stack Development. Elsevier Procedia published researcher (XGBoost: 93.78% accuracy) and 2x hackathon winner; proficient in Python, TensorFlow, OpenCV, Scikit-Learn, Django, Flask, REST API design, MySQL, PostgreSQL, Docker, and Agile/Scrum workflows. Available for ML Engineer / SDE internship (2025) or full-time roles (May 2027).

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C, HTML5, CSS3

AI / ML / DL: Machine Learning, Deep Learning, Neural Networks, Computer Vision, NLP, TensorFlow, OpenCV, Scikit-Learn, XGBoost, Random Forest, Logistic Regression, Pandas, NumPy, Matplotlib, Feature Engineering

Generative AI: Vertex AI, LLM Integration, Prompt Engineering, Google Gemini API

Web & Backend: Django, Flask, Next.js, React.js, Redux, Node.js, REST API Design, JWT Authentication, Firebase, Firestore

Databases: MySQL, PostgreSQL, Firebase Firestore, Database Design, Query Optimization

Cloud & DevOps: AWS (S3, EC2), GCP, Docker, GitHub Actions (CI/CD), Git, Vercel, Linux

Testing: Pytest, Unit Testing, API Testing, Test-Driven Development (TDD)

CS Fundamentals: Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Networks, Agile/Scrum, SDLC

Education

B.Tech – Artificial Intelligence & Machine Learning

2023 – 2027

United University, Prayagraj

CGPA: 8.0 / 10

Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures & Algorithms, System Design, DBMS, Operating Systems, Computer Networks

Work Experience

Backend Development Intern

Aug 2025 – Present

Prodesk IT, Noida

- Engineered 8+ REST API endpoints in Python/Django for R&D production apps, reducing API response latency by 30% through query optimization and backend logic refactoring.
- Designed scalable MySQL database workflows, cutting backend response time by 25%; collaborated in Agile/Scrum team of 5+ engineers across 2 production releases.
- Resolved 15+ production bugs and wrote Pytest unit tests for 3 core API modules, improving overall application stability and test coverage by 40%.

Web Developer Intern

Jun 2025 – Jul 2025

Coderesite, Prayagraj

- Built and deployed 3+ responsive full-stack web applications using HTML5, CSS3, JavaScript, and Firebase backend, improving page load speed by 35%.
- Reduced UI bugs by 40% through systematic testing and code refactoring; all projects deployed via Vercel with CI/CD pipelines.

Freelance Full Stack Developer

Jan 2024 – Mar 2025

Hanuman Pushpa Varsha Committee, Prayagraj

- Built and deployed full-stack bilingual (Hindi-English) platform using Next.js, Firebase, and Vercel with 6 modules: Razorpay payment gateway, live streaming, member registry, and admin dashboard; currently serving 500+ active users.
- Reduced admin overhead by 60% by automating member registration via Firebase Firestore real-time database with role-based access control (RBAC).

Projects

SignSetu – Real-Time AI Bidirectional Sign Language Platform | [GitHub](#) [Live](#)

- Python, TensorFlow, OpenCV, MediaPipe, Flask, React.js, Firebase, Docker, Vercel, CI/CD
- Winner, UHack 4.0** (1st/500+ participants): built real-time ISL gesture recognition system using TensorFlow CNN + MediaPipe; achieved **91%+ classification accuracy at 24 fps** across 26 sign classes.
- Curated 5,000+ sample custom ISL dataset using feature engineering and data augmentation; deployed with JWT authentication, REST API, Firebase backend, Docker, and Vercel CI/CD.

QuickSeva – LLM-Powered AI Civic Services Assistant | [GitHub](#) [Live](#)

- Python, React.js, Google Gemini API, LLM Integration, Prompt Engineering, Flask, Firebase, Vercel
- Winner, Google Build with AI Hackathon** (GDG Prayagraj): LLM-powered assistant using Google Gemini API to help citizens navigate government schemes via conversational UI with real-time streaming responses.
- Designed REST API backend with Firebase Firestore session management; applied prompt engineering techniques for accurate scheme retrieval and response quality.

Aarogyam – Mental Wellness Platform | [GitHub](#) [Live](#)

- JavaScript, HTML5, CSS3, Firebase Authentication, Firestore, REST API
- Developed open-source wellness platform with 9+ therapy modules and personalized content delivery; implemented Firebase Auth and cloud database integration for 100+ active users.

Research Publication

Multi-Class Anomaly Detection in Network Traffic Using Supervised Machine Learning

- **Published** – Elsevier Procedia Computer Science (ICMLDE 2025); presented at AICCT-2025 International Conference, Sept 2025.
- Benchmarked Decision Tree, Random Forest, Logistic Regression, and XGBoost on CIC UNSW-NB15 dataset (175,000+ records); **XGBoost achieved 93.78% accuracy with Precision, Recall, F1, ROC-AUC**; ensemble methods outperformed linear models on rare attack classes.

Achievements & Leadership

Winner – UHack 4.0 (1st/500+ participants): SignSetu – AI real-time sign language translation platform.

Winner – Google Build with AI Hackathon (GDG Prayagraj): QuickSeva recognized by Google Developer ecosystem judges using Vertex AI and Google Gemini API.

Research Published – Elsevier Procedia Computer Science (ICMLDE 2025); Certificate of Appreciation, AICCT-2025.

Top 10 – BuildX Hackathon, NSUT New Delhi (national level); **Top 20** – FlutterFlow Hackathon, IIIT Allahabad.

Co-Lead Organizer – HackDiwas 3.0 State-Level Hackathon (500+ participants, 48 hrs); **TEDx United University Organizer**: speaker outreach and technical operations.

Community: FOSS Prayagraj (Design & Media), GDG Prayagraj DevFest 2025 (GenAI, Vertex AI), WikiClub UU (Social Media Head).